

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10							***** VALVES *****			810	NIPPLE TO BE CUT FROM SCH 160 PIPE.
20		VA5					***** MODULAR VALVES *****			817	USE ONLY FOR TIE IN
30										821	TO BE USED FOR LINES SUBJECT TO AIV OR WHERE REQUIRED BY STRESS.
30	U04		MODULAR DBB	K01AC02	0.5	2	MODULAR VALVE DB&B BALL TYPE 150# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	827	SUBSTITUTED MATERIAL BY FABRICATOR - USE ONLY WITH CONSENT OF MATERIALS ENGINEER.
40	U04	VAV	MODULAR DBB	K03AC02	3	3	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	B1	ALL BOLTS ARE METRIC DIA.
50	U04		MODULAR DBB	M01AC02	0.5	2	MODULAR VALVE DB&B BALL TYPE 150# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO	0	0	B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10
60	U04	FS1	MODULAR DBB	M03AC02	0.75	3	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	B6	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AS22
70	U04	ZZ6	MODULAR DBB	M06AC02	0.75	1.5	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.
80	U04	VAW	MODULAR DBB	M03EC02	0.75	1	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	1.5	3	FS1	USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.
90		VB2					***** BALL VALVES *****			GA6	SPECIAL GASKET ONLY FOR USE AS SPECIFIED IN ENG-200-117
90										NDA	USE ONLY WITH CONSENT OF MATERIALS ENGINEER. USE THIS VALVE TO SUBSTITUTE CLASS 300 WITH CLASS 600 FOR CLOSED DRAIN FLUSHING RING OUTLET CASES ONLY.
100										PJ9	FOR USE IN NON-JACKETED SULPHUR LINES ONLY.
100	U04		BALL	A01AC01	3	4	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT RB LO	0	0	SB6	USE JACKSCREWS FOR PADDLE
110	U04		BALL	A01AC02	3	4	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL RB LO	0	0		BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.
120	U04		BALL	A01AC03	6	6	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT RB GO	0	0	SC1	USE 3/4 IN SCREWED FLANGES FOR SAMPLE SUPPLY AND RETURN CONNECTIONS AND 1/2 IN 8amp; 3/4 IN SCREWED FLANGES STEAM SUPPLY/CONDENSATE RETURN LINES FOR CONTRATRACE AND ORIFICE SETS
130	U04		BALL	A01AC04	6	6	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL RB GO	0	0	UD6	USE FOR UNVALVED CAPPED HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST.
140	U04		BALL	A01AC07	8	8	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT RB GO	0	0	UD7	USE THREDOLET AND SCREWED PLUG FOR HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST
150	U04		BALL	A01AC08	8	8	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL RB GO	0	0	UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.
160	U04		BALL	B01AC01	0.75	2	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO	0	0	UJ4	FOR 2 INCH THERMOWELL CONNECTIONS ONLY
170	U04	VA5	BALL	B01AC01	3	3	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO	0	0	UK9	DO NOT USE FOR ORIFICE FLANGE TAP ASSEMBLIES.
180	U04		BALL	B01AC02	0.75	2	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB LO	0	0	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
190	U04	VA5	BALL	B01AC02	3	3	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB LO	0	0	VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.
200	U04	VA5	BALL	B01AC03	4	4	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB GO	0	0	VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.
210	U04	VA5	BALL	B01AC03	4	4	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO	0	0	VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.
220	U04	VA5	BALL	B01AC04	4	4	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB GO	0	0	VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.
230	U04	VA5	BALL	B01AC04	6	6	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB LO	0	0	VQ2	QUARTER TURN VALVES ARE TO BE SELECTED FOR PROCESS APPLICATIONS.
240	U04	VA5	BALL	B01AC07	6	8	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO	0	0	VQ3	TO BE USED FOR COMMISSIONING ONLY
250	U04	VA5	BALL	B01AC07	10	24	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO	0	0	ZZ6	USE ONLY WHEN REQUIREMENT SHOWN IN PID.
260	U04	VA5	BALL	B01AC08	6	24	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB GO	0	0		
270	U04	VA5	BALL	B01JC07	26	56	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO B16.47 SERIES A	0	0		
280	U04	VA5	BALL	B01JC08	26	56	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB GO B16.47 SERIES A	0	0		
290	U04	FS1	BALL	B03AC01	0.75	2	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO	0	0		
300	U04	FS1	BALL	B03AC02	0.75	2	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB LO	0	0		
310	U04	FS1	BALL	A03AC01	3	3	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO	0	0		
320	U04	FS1	BALL	A03AC02	3	3	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL RB LO	0	0		
330	U04	NDA		B06AC02	0.75	0.75	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB LO	0	0		
340							***** RISING STEM BALL VALVES *****				
340											
350											
350			BALL	R01AC06	0.75	10	BALL RISING STEM 150# RF LTCS A352 LCC 316SS TRIM METAL RB HO	0	0		
360							***** BUTTERFLY VALVES *****				
360											
370											
370			BUTTERFLY	T01AC61	10	24	BUTTERFLY 150# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO	0	0		
380			BUTTERFLY	T01JC61	26	56	BUTTERFLY 150# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO ASME B16.47 SERIES A	0	0		
390							***** GATE VALVES *****				
390		VQ2									
400											
400	U04		GATE	G01AC48	0.75	1.5	GATE 150# RF LTCS A350 LF2 CL.1 TRIM 16 WEDGE HW	0	0		
410	U04		GATE	G01AC41	2	8	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW	0	0		
420			GATE	G01AC42	10	20	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE GO	0	0		
430		VQ3	GATE	G01JC42	42	42	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE GO B16.47 SERIES A	0	0		
440			GATE	G03AC41	2	2	GATE 300# RF LTCS A352 LCC TRIM 16 WEDGE HW	0	0		
450							***** THROUGH CONDUIT GATE VALVES *****				
450		VQ2									
460	U04		GATE	W01AC70	3	16	GATE 150# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW	0	0		
470	U04		GATE	W01AC71	18	24	GATE 150# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO	0	0		
480			GATE	W01JC71	26	56	GATE 150# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO B16.47 SERIES A	0	0		
490							***** GLOBE VALVES *****				
490											
500											
500	U04		GLOBE	L01AC48	0.75	1.5	GLOBE 150# RF LTCS A350 LF2 CL.1 TRIM 16 OS&Y HW	0	0		
510			GLOBE	L01AC41	2	6	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y HW	0	0		
520	U04		GLOBE	L01AC42	8	12	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y GO	0	0		
530	U04			L03AC41	6	6	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW	0	0		
540											

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - ALL SIZES. SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL:LTCS FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 232 °C COMMENT: REFER NOTE 800	12/16/2024	BY OZAC	U04	Page 1 of 7	150K21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
540							***** CHECK VALVES *****			
550										
550	U04	VC1	CHECK	C01AC50	0.75	1.5	CHECK PISTON 150# RF LTCS A350 LF2 CL.1 TRIM 16	0	0	
560	U04		CHECK	C01AC45	2	6	CHECK DUAL WAFER LUG 150# RF LTCS A352 LCC TRIM 16 THROU BOLTED	0	0	
570	U04		CHECK	C01AC52	8	24	CHECK DUAL DBL FLG 150# RF LTCS A352 LCC TRIM 16	0	0	
580	U04		CHECK	C01JC52	26	56	CHECK DUAL DBL FLG 150# RF LTCS A352 LCC TRIM 16 B16.47 SERIES A	0	0	
590										
590							***** PIPE *****			
600										
600	U04		PIPE	5358508	0.5	1.5	PIPE SCH XS SMLS LTCS A333 GR.6 PE	0	0	
610	U04	821	PIPE	5358571	6	6	PIPE SCH 120 SMLS LTCS A333 GR.6 BE	0	0	
620	U04	821	PIPE	5358515	2	2	PIPE SCH 160 SMLS LTCS A333 GR.6 BE	0	0	
630			PIPE	5358514	2	12	PIPE SCH XS SMLS LTCS A333 GR.6 BE	0	0	
640			PIPE	5358505	3	14	PIPE SCH STD SMLS LTCS A333 GR.6 BE	0	0	
650			PIPE	5358501	16	16	PIPE SCH 20 SMLS LTCS A333 GR.6 BE	0	0	
660			PIPE	5358597	18	24	PIPE SCH 20 EFW LTCS A671 CC60 CL.22 BE	0	0	
670	U04	827	PIPE	5358506	20	20	PIPE SCH 60 SMLS LTCS A333 GR.6 BE	0	0	
680	U04	821	PIPE	5358555	20	20	PIPE SCH XS EFW LTCS A671 CC60 CL.22 BE	0	0	
690			PIPE	5358577	18	44	PIPE SCH XS EFW LTCS A671 CC60 CL.22 BE	0	0	
700	U04	821	PIPE	5358385	18	18	PIPE SCH 30 EFW LTCS A671 CC60 CL.22 BE	0	0	
710	U04	821	PIPE	5358556	20	20	PIPE SCH 40 EFW LTCS A671 CC60 CL.22 BE	0	0	
720	U04	821	PIPE	5358580	24	24	PIPE SCH 40 EFW LTCS A671 CC60 CL.22 BE	0	0	
730	U04	821	PIPE	5358613	36	36	PIPE SCH 40 EFW LTCS A671 CC60 CL.22 BE	0	0	
740			PIPE	5358540	46	48	PIPE WT 14.0MM EFW LTCS A671 CC60 CL.22 BE	0	0	
750			PIPE	5358610	56	56	PIPE WT 15.0MM EFW LTCS A671 CC60 CL.22 BE	0	0	
760		810	PIPE	5358512	0.5	0.5	PIPE SCH 160 SMLS LTCS A333 GR.6 PE	0	0	
770	U04	UK9	PIPE	5358511	0.5	0.75	PIPE SCH XXS SMLS LTCS A333 GR.6 PE	0	0	
780	U04		PIPE	5358747	18	42	PIPE WT 19.05MM EFW LTCS A671 CC60 CL.22 BE	0	0	
790	U04	821	PIPE	5358545	14	16	PIPE WT 19.05MM SMLS LTCS A333 GR.6 BE	0	0	
800	U04	821	PIPE	5358544	36	36	PIPE WT 22.2MM EFW LTCS A671 CC60 CL.22 BE	0	0	
810	U04	821	PIPE	5358546	48	48	PIPE WT 25.4MM EFW LTCS A671 CC60 CL.22 BE	0	0	
820										
820							***** FITTINGS *****			
830										
830	U04		ELBOW	5392600	0.5	2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6	0	0	
840	U04	821	ELBOW	5392816	2	2	ELL BW 90 DEG LR SCH 160 LTCS A420 WPL6	0	0	
850	U04	UK9	ELBOW	5392815	0.5	0.75	ELL BW 90 DEG LR SCH XXS LTCS A420 WPL6	0	0	
860			ELBOW	5392438	3	14	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6	0	0	
870			ELBOW	5392276	16	16	ELL BW 90 DEG LR SCH 20 LTCS A420 WPL6	0	0	
880			ELBOW	5392280	18	24	ELL BW 90 DEG LR SCH 20 LTCS A420 WPL6-W	0	0	
890			ELBOW	5392497	18	44	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6-W	0	0	
900	U04	821	ELBOW	5392700	6	6	ELL 90 DEG LR SCH 120 LTCS A420 WPL6	0	0	
910	U04	821	ELBOW	5392110	18	18	ELL BW 90 DEG LR SCH 30 LTCS A420 WPL6-W	0	0	
920	U04	821	ELBOW	5392111	20	24	ELL BW 90 DEG LR SCH 40 LTCS A420 WPL6-W	0	0	
930			ELBOW	5392472	46	48	ELL BW 90 DEG LR WT 14.0MM LTCS A420 WPL6-W	0	0	
940			ELBOW	5392499	56	56	ELL 90 DEG LR WT 15.0MM LTCS A420 WPL6-W	0	0	
950	U04	821	ELBOW	5392442	14	14	ELL BW 90 DEG LR WT 19.05 MM LTCS A420 WPL6	0	0	
960			ELBOW	5392489	20	42	ELL BW 90 DEG LR WT 19.05 MM LTCS A420 WPL6-W	0	0	
970	U04	821	ELBOW	5392266	36	36	ELL BW 90 DEG LR WT 22.2MM LTCS A420 WPL6-W	0	0	
980	U04	821	ELBOW	5392967	48	48	ELL BW 90 DEG LR WT 25.4MM LTCS A420 WPL6-W	0	0	
990	U04		ELBOW	5392606	0.5	2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6	0	0	
1000			ELBOW	5392444	3	14	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6	0	0	
1010			ELBOW	5392282	16	16	ELL BW 45 DEG SCH 20 LTCS A420 WPL6	0	0	
1020			ELBOW	5392281	18	24	ELL BW 45 DEG LR SCH 20 LTCS A420 WPL6-W	0	0	
1030			ELBOW	5392608	24	44	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6-W	0	0	
1040	U04	821	ELBOW	5392701	6	6	ELL BW 45 DEG LR SCH 120 LTCS A420 WPL6	0	0	
1050	U04	UK9	ELBOW	5392831	0.5	0.75	ELL BW 45 DEG LR SCH XXS LTCS A420 WPL6	0	0	
1060			ELBOW	5392590	46	48	ELL BW 45 DEG LR WT 14.0MM LTCS A420 WPL6-W	0	0	
1070			ELBOW	5392520	56	56	ELL BW 45 DEG LR WT 15.0MM LTCS A420 WPL6 -W	0	0	
1080	U04	821	ELBOW	5392267	36	36	ELL BW 45 DEG LR WT 22.2MM LTCS A420 WPL6 -W	0	0	
1090	U04	821	ELBOW	5392968	48	48	ELL BW 45 DEG LR WT 25.4MM LTCS A420 WPL6-W	0	0	
1100										
1100	U04		TEE	5392609	0.5	2	TEE BW SCH XS LTCS A420 WPL6	0	0	
1110	U04	UK9	TEE	5392879	0.5	0.75	TEE SCH XXS LTCS A420 WPL6	0	0	
1120			TEE	5392447	3	14	TEE BW SCH STD LTCS A420 WPL6	0	0	
1130			TEE	5392285	16	16	TEE BW SCH 20 LTCS A420 WPL6	0	0	
1140			TEE	5392295	18	24	TEE BW SCH 20 LTCS A420 WPL6-W	0	0	
1150	U04	821	TEE	5392412	18	18	TEE SCH 30 LTCS A420 WPL6-W	0	0	
1160			TEE	5392406	26	44	TEE BW SCH XS LTCS A420 WPL6-W	0	0	
1170	U04		TEE	5392253	30	42	TEE BW WT 19.05MM LTCS A420 WPL6-W	0	0	
1180	U04	821	TEE	5392252	48	48	TEE BW WT 25.4MM LTCS A420 WPL6-W	0	0	
1190	U04		TEE	5392251	46	48	TEE BW WT 14.0MM LTCS A420 WPL6-W	0	0	
1200			TEE	5392407	56	56	TEE BW WT 15.0MM LTCS A420 WPL6-W	0	0	

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - ALL SIZES. SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL:LTCS FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 232 °C COMMENT: REFER NOTE 800	12/16/2024	BY OZAC	U04	Page 2 of 7	150K21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1210			TEE RED	5392661	0.75	1.5	TEE RED BW SCH XS LTCS A420 WPL6	1	2	
1220			TEE RED	5392454	1.5	2	TEE RED BW SCH STD/XS LTCS A420 WPL6	3	4	
1230			TEE RED	5392445	3	12	TEE RED BW SCH STD LTCS A420 WPL6	4	14	
1240			TEE RED	5392289	6	14	TEE RED BW SCH 20/STD LTCS A420 WPL6	16	16	
1250			TEE RED	5392296	8	14	TEE RED BW SCH 20/STD LTCS A420 WPL6-W	18	24	
1260			TEE RED	5392476	14	14	TEE RED SCH XS/STD LTCS A420 WPL6-W	26	32	
1270			TEE RED	5392292	16	20	TEE RED BW SCH 20 LTCS A420 WPL6-W	18	24	
1280			TEE RED	5392475	16	24	TEE RED BW SCH XS/20 LTCS A420 WPL6-W	26	42	
1290			TEE RED	5392413	26	42	TEE RED SCH XS LTCS A420 WPL6-W	28	44	
1300			TEE RED	5392650	26	42	TEE RED WT 14.0MM/SCH XS LTCS A420 WPL6-W	46	48	
1310			TEE RED	5392653	24	24	TEE RED WT 14.0MM/SCH 20 LTCS A420 WPL6-W	48	48	
1320			TEE RED	5392321	42	44	TEE RED BW WT 15.0MM/XS LTCS A420 WPL6-W	56	56	
1330			TEE RED	5392651	48	48	TEE RED BW WT 15.0MM/14.0MM LTCS A420 WPL6-W	56	56	
1340	U04	821	TEE RED	5392443	18	24	TEE RED BW 19.05MM/20 LTCS A420 WPL6-W	30	30	
1350	U04	821	TEE RED	5392346	10	10	TEE RED BW 19.05MM/STD LTCS A420 WPL6-W	20	20	
1360	U04	821	TEE RED	5392436	20	20	TEE RED BW 19.05MM/XS LTCS A420 WPL6-W	30	30	
1370	U04	821	TEE RED	5392440	14	14	TEE RED BW WT 19.05MM LTCS A420 WPL6-W	30	30	
1380	U04		TEE RED	5392679	16	20	TEE RED SCH 40/SCH 20 LTCS A420 WPL6-W	24	36	
1390	U04	821	TEE RED	5392678	20	20	TEE RED SCH 40/SCH 20 LTCS A420 WPL6-W	36	36	
1400	U04	821	TEE RED	5392677	24	24	TEE RED SCH 40/SCH XS LTCS A420 WPL6-W	36	36	
1410	U04	821	TEE RED	5392401	10	10	TEE RED SCH 40/STD LTCS A420 WPL6-W	24	24	
1420	U04	821	TEE RED	5392680	20	20	TEE RED SCH XS/20 LTCS A420 WPL6-W	30	30	
1430	U04	821	TEE RED	5392676	16	16	TEE RED WT 22.2MM/SCH 20 LTCS A420 WPL6-W	36	36	
1440	U04	821	TEE RED	5392675	24	24	TEE RED WT 22.2MM/SCH XS LTCS A420 WPL6-W	36	36	
1450	U04	821	TEE RED	5392674	42	42	TEE RED WT 25.4MM/SCH XS LTCS A420 WPL6-W	48	48	
1460										
1460			CAP	5392610	0.75	2	CAP BW SCH XS LTCS A420 WPL6	0	0	
1470			CAP	5392610	0.75	44	CAP BW SCH XS LTCS A420 WPL6	0	0	
1480			CAP	5392448	3	14	CAP BW SCH STD LTCS A420 WPL6	0	0	
1490			CAP	5392286	16	24	CAP BW SCH 20 LTCS A420 WPL6	0	0	
1500			CAP	5392652	46	48	CAP BW WT 14.0MM LTCS A420 WPL6	0	0	
1510			CAP	5392615	56	56	CAP BW WT 15.0MM LTCS STL A420 WPL6	0	0	
1520	U04	821	CAP	5392669	36	36	CAP BW SCH 40 LTCS A420 WPL6	0	0	
1530	U04	821	CAP	5392744	30	30	CAP BW WT 19.05MM LTCS A420 WPL6	0	0	
1540	U04	821	CAP	5392745	36	36	CAP BW WT 22.2MM LTCS A420 WPL6	0	0	
1550	U04	821	CAP	5392746	48	48	CAP BW WT 25.4MM LTCS A420 WPL6	0	0	
1560										
1560			RED CONC	5392458	0.5	1.5	REDUCER CONC SCH XS LTCS A420 WPL6	0.75	2	
1570			RED CONC	5392452	1.5	2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3	4	
1580			RED CONC	5392449	3	12	REDUCER CONC SCH STD LTCS A420 WPL6	4	14	
1590	U04	821	RED CONC	5392704	4	4	REDUCER CONC SCH 120 LTCS A420 WPL6	6	6	
1600	U04	821	RED CONC	5392467	2	2	REDUCER CONC SCH 120/160 LTCS A420 WPL6	4	4	
1610			RED CONC	5392290	8	14	REDUCER CONC SCH 20/STD LTCS A420 WPL6	16	16	
1620			RED CONC	5392297	10	14	REDUCER CONC SCH 20/STD LTCS A420 WPL6-W	18	20	
1630			RED CONC	5392287	16	20	REDUCER CONC SCH 20 LTCS A420 WPL6-W	18	24	
1640			RED CONC	5392512	18	24	REDUCER CONC SCH XS/20 LTCS A420 WPL6-W	24	30	
1650			RED CONC	5392507	26	42	REDUCER CONC SCH XS LTCS A420 WPL6-W	28	44	
1660			RED CONC	5392260	40	44	REDUCER CONC WT 14.0MM/ SCH XS LTCS A420 WPL6-W	46	48	
1670	U04	821	RED CONC	5392976	36	36	REDUCER CONC SCH XS/WT 22.2MM LTCS A420 WPL6-W	40	40	
1680			RED CONC	5392315	42	44	REDUCER CONC WT 15.0MM/ XS LTCS A420 WPL6-W	56	56	
1690			RED CONC	5392261	46	48	REDUCER CONC WT 15.0MM/ 14.0MM LTCS A420 WPL6-W	56	56	
1700	U04		RED CONC	5392963	18	24	REDUCER CONC WT 19.05MM LTCS A420 WPL6-W	24	30	
1710	U04	821	RED CONC	5392962	18	20	REDUCER CONC WT 19.05MM/SCH 20 LTCS A420 WPL6-W	24	30	
1720	U04	821	RED CONC	5392961	14	14	REDUCER CONC WT 19.05MM/SCH STD LTCS A420 WPL6-W	20	20	
1730	U04	821	RED CONC	5392973	40	42	REDUCER CONC WT 25.4MM/ SCH XS LTCS A420 WPL6-W	48	48	
1740			RED ECC	5392461	0.5	1.5	REDUCER ECC SCH XS LTCS A420 WPL6	0.75	2	
1750			RED ECC	5392453	1.5	2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3	4	
1760			RED ECC	5392450	3	12	REDUCER ECC SCH STD LTCS A420 WPL6	4	14	
1770			RED ECC	5392291	8	14	REDUCER ECC SCH 20/STD LTCS A420 WPL6	16	16	
1780	U04	821	RED ECC	5392705	4	4	REDUCER ECC SCH 120 LTCS A420 WPL6	6	6	
1790	U04	821	RED ECC	5392468	2	2	REDUCER ECC SCH 120/160 LTCS A420 WPL6	4	4	
1800	U04	821	RED ECC	5392271	8	8	REDUCER ECC SCH 20/XS LTCS A420 WPL6	16	16	
1810	U04	821	RED ECC	5392279	24	24	REDUCER ECC SCH 40/SCH 30 LTCS A420 WPL6-W	36	36	
1820	U04		RED ECC	5392612	6	12	REDUCER ECC SCH XS LTCS A420 WPL6	10	20	
1830	U04	821	RED ECC	5392459	6	6	REDUCER ECC SCH XS/STD LTCS A420 WPL6	12	12	
1840			RED ECC	5392298	10	14	REDUCER ECC SCH 20/STD LTCS A420 WPL6-W	18	20	
1850			RED ECC	5392288	16	20	REDUCER ECC SCH 20 LTCS A420 WPL6-W	18	24	
1860			RED ECC	5392513	18	24	REDUCER ECC SCH XS/20 LTCS A420 WPL6-W	24	36	
1870	U04	821	RED ECC	5392278	18	18	REDUCER ECC SCH 30 LTCS A420 WPL6-W	24	24	
1880	U04	821	RED ECC	5392283	10	12	REDUCER ECC SCH 30/STD LTCS A420 WPL6-W	18	18	
1890	U04	821	RED ECC	5392396	20	20	REDUCER ECC SCH 40 LTCS A420 WPL6-W	24	24	
1900	U04		RED ECC	5392589	20	24	REDUCER ECC SCH 40/SCH 20 LTCS A420 WPL6-W	24	36	
1910			RED ECC	5392514	10	42	REDUCER ECC SCH XS LTCS A420 WPL6-W	18	44	

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - ALL SIZES. SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL:LTCS FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 232 °C COMMENT: REFER NOTE 800	12/16/2024	BY OZAC	U04	Page 3 of 7	150K21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1920	U04		RED ECC	5392623	20	30	REDUCER ECC 19.05MM LTCS A420 WPL6-W	30	42	
1930			RED ECC	5392262	40	44	REDUCER ECC WT 14.0MM/SCH XS LTCS A420 WPL6-W	46	48	
1940			RED ECC	5392515	42	44	REDUCER ECC 15.0MM/ XS LTCS A420 WPL6-W	56	56	
1950			RED ECC	5392263	46	48	REDUCER ECC WT 15.0MM/14.0MM LTCS A420 WPL6-W	56	56	
1960	U04	821	RED ECC	5392977	36	36	REDUCER ECC SCH XS/ WT 22.2MM LTCS A420 WPL6-W	40	40	
1970	U04		RED ECC	5392510	12	14	REDUCER ECC SCH XS/STD LTCS A420 WPL6-W	18	20	
1980	U04	821	RED ECC	5392471	8	8	REDUCER ECC WT 19.05 MM/STD LTCS A420 WPL6	16	16	
1990	U04	821	RED ECC	5392446	20	20	REDUCER ECC WT 19.05MM / SCH 20 LTCS A420 WPL6-W	24	30	
2000	U04	821	RED ECC	5392385	12	12	REDUCER ECC WT 19.05MM / STD LTCS A420 WPL6-W	20	20	
2010	U04	821	RED ECC	5392974	40	42	REDUCER ECC WT 25.4MM/SCH XS LTCS A420 WPL6-W	48	48	
2020										
2020	U04		OLETS	5504001	0.75	1.5	NIPOFLANGE 150# RF SCH XS LTCS A350 LF2 CL.1	2	2	
2030	U04		OLETS	5504001	0.75	1.5	NIPOFLANGE 150# RF SCH XS LTCS A350 LF2 CL.1	26	44	
2040	U04		OLETS	5504001	2	2	NIPOFLANGE 150# RF SCH XS LTCS A350 LF2 CL.1	26	44	
2050	U04		OLETS	5504220	0.75	0.75	NIPOFLANGE 1500# RF SCH STD/XS LTCS A350 LF2 CL.1	8	8	
2060	U04		OLETS	5504002	0.75	1.5	NIPOFLANGE 150# RF SCH STD/ XS LTCS A350 LF2 CL.1	3	14	
2070	U04		OLETS	5504002	2	2	NIPOFLANGE 150# RF SCH STD/ XS LTCS A350 LF2 CL.1	3	14	
2080	U04		OLETS	5504016	0.75	1.5	NIPOFLANGE 150# RF SCH 20/XS LTCS A350 LF2 CL.1	16	24	
2090	U04		OLETS	5504016	2	2	NIPOFLANGE 150# RF SCH 20/XS LTCS A350 LF2 CL.1	16	24	
2100	U04	UJ2	OLETS	5504157	0.75	2	NIPOFLANGE 150# RF WT 14.0MM/SCH XS LTCS A350 LF2 CL.1	46	48	
2110	U04		OLETS	5504100	0.75	2	NIPOFLANGE 150# RF WT 15.0MM/XS LTCS A350 LF2 CL.1	56	56	
2120	U04	UJ2	OLETS	5504020	0.75	1.5	NIPOFLANGE 300# RF SCH XS LTCS A350 LF2 CL.1	1.5	30	
2130	U04	UJ4	OLETS	5504222	2	2	NIPOFLANGE 300# RF SCH 20/160 LTCS A350 LF2 CL.1	24	24	
2140	U04	UJ2	OLETS	5504017	0.75	1.5	NIPOFLANGE 300# RF SCH 20/XS LTCS A350 LF2 CL.1	16	24	
2150	U04		OLETS	5504326	2	2	NIPOFLANGE 300# RF SCH STD/160 LTCS A350 LF2 CL.1	6	14	
2160	U04	UJ4	OLETS	5504064	2	2	NIPOFLANGE 300# RF SCH XS LTCS A350 LF2 CL.1	36	36	
2170	U04		OLETS	5504323	2	4	NIPOFLANGE 300# RF SCH XS/160 LTCS A350 LF2 CL.1	30	30	
2180	U04			5504423	1.5	1.5	NIPOFLANGE 300# RF WT 14.0MM/SCH XS LTCS A350 LF2 CL.1	48	48	
2190	U04		OLETS	5504021	0.75	1.5	NIPOFLANGE 300# RF SCH STD/XS LTCS A350 LF2 CL.1	3	10	
2200	U04	UJ2	OLETS	5504025	0.75	1.5	NIPOFLANGE 300# RF SCH STD/160 LTCS A350 LF2 CL.1	4	14	
2210	U04	UJ2	OLETS	5504019	0.75	1.5	NIPOFLANGE 300# RF SCH 20/160 LTCS A350 LF2 CL.1	16	24	
2220	U04	UJ2	OLETS	5504023	0.75	1.5	NIPOFLANGE 300# RF SCH XS/160 LTCS A350 LF2 CL.1	26	44	
2230	U04	UJ2	OLETS	5504158	0.75	1.5	NIPOFLANGE 300# RF WT 14.0MM/160 LTCS A350 LF2 CL.1	46	48	
2240	U04	UJ2	OLETS	5504112	0.75	1.5	NIPOFLANGE 300# RF WT 15.0MM/160 LTCS A350 LF2 CL.1	56	56	
2250	U04		OLETS	5504007	0.75	0.75	NIPOFLANGE 600# RF SCH STD/XS LTCS A350 LF2 CL.1	6	8	
2260	U04	UJ2	OLETS	5504006	0.75	1.5	NIPOFLANGE 600# RF SCH XS LTCS A350 LF2 CL.1	2	36	
2270	U04		OLETS	5504500	0.75	0.75	NIPOFLANGE FLAT BOTTOM 150# RF SCH XS LTCS A350 LF2 CL.1	0	0	
2280	U04		OLETS	5504501	0.75	0.75	NIPOFLANGE FLAT BOTTOM 300# RF SCH XS LTCS A350 LF2 CL.1	0	0	
2290										
2290	U04		OLETS	5505335	0.75	4	WELDOLET SCH XS LTCS A350 LF2 CL.1	2	44	
2300	U04	821	OLETS	5505552	6	6	WELDOLET WT 22.2MM/STD LTCS A350 LF2 CL.1	36	36	
2310	U04	821	OLETS	5505563	1	2	WELDOLET WT 25.4MM/SCH XS LTCS A350 LF2 CL.1	48	48	
2320	U04		OLETS	5505179	0.75	2	WELDOLET SCH STD/XS LTCS A350 LF2 CL.1	3	14	
2330	U04		OLETS	5505572	0.75	2	WELDOLET SCH 20/XS LTCS A350 LF2 CL.1	16	24	
2340	U04		OLETS	5505560	0.75	2	WELDOLET WT 14.0MM/SCH XS LTCS A350 LF2 CL.1	46	48	
2350	U04		OLETS	5505575	0.75	2	WELDOLET WT 15.0MM/XS LTCS A350 LF2 CL.1	56	56	
2360	U04		OLETS	5505164	3	4	WELDOLET SCH STD LTCS A350 LF2 CL.1	6	14	
2370	U04		OLETS	5505571	3	8	WELDOLET SCH 20/STD LTCS A350 LF2 CL.1	16	24	
2380	U04		OLETS	5505334	3	12	WELDOLET SCH XS/STD LTCS A350 LF2 CL.1	26	44	
2390	U04		OLETS	5505561	3	12	WELDOLET WT 14.0MM/STD LTCS A350 LF2 CL.1	46	48	
2400	U04		OLETS	5505576	3	6	WELDOLET WT 15.0MM/STD LTCS A350 LF2 CL.1	56	56	
2410	U04		OLETS	5504376	3	4	WELDOLET FLAT BOTTOM SCH STD LTCS A350 LF2 CL.1	0	0	
2420	U04		OLETS	5505562	4	12	WELDOLET WT 25.4MM/STD LTCS A350 LF2 CL.1	48	48	
2430	U04		OLETS	5504374	0.75	2	WELDOLET FLAT BOTTOM SCH XS LTCS A350 LF2 CL.1	0	0	
2440										
2440	U04		OLETS	5509731	3	14	SWEEPOLET 19.05 MM/STD LTCS A350 LF2 CL.1	16	42	
2450	U04		OLETS	5509740	3	6	SWEEPOLET SCH 20/STD LTCS A350 LF2 CL.1	16	24	
2460	U04		OLETS	5509720	1.5	2	SWEEPOLET SCH 20/XS LTCS A350 LF2 CL.1	16	24	
2470	U04	821	OLETS	5509741	3	4	SWEEPOLET SCH 30/STD LTCS A350 LF2 CL.1	18	18	
2480	U04	821	OLETS	5509742	2	2	SWEEPOLET SCH 30/XS LTCS A350 LF2 CL.1	18	18	
2490	U04	821	OLETS	5509719	3	12	SWEEPOLET SCH 40/STD LTCS A350 LF2 CL.1	36	36	
2500	U04	821	OLETS	5509732	2	2	SWEEPOLET SCH 40/XS LTCS A350 LF2 CL.1	24	36	
2510	U04	821	OLETS	5509734	3	3	SWEEPOLET SCH STD LTCS A350 LF2 CL.1	10	14	
2520	U04		OLETS	5509730	2	2	SWEEPOLET SCH XS LTCS A350 LF2 CL.1	8	30	
2530	U04		OLETS	5509729	3	6	SWEEPOLET SCH XS/STD LTCS A350 LF2 CL.1	18	30	
2540	U04	821	OLETS	5509724	2	2	SWEEPOLET STD/XS LTCS A350 LF2 CL.1	10	12	
2550	U04		OLETS	5509728	2	2	SWEEPOLET WT 19.05 MM/XS LTCS A350 LF2 CL.1	16	30	
2560	U04	821	OLETS	5509727	3	12	SWEEPOLET XS/STD LTCS A350 LF2 CL.1	36	36	
2570										
2570	U04	UJ2	ELBOFLG	5516625	1.5	1.5	ELBOFLANGE 300# RF SCH STD/160 LTCS A350 LF2 CL.1	4	14	
2580	U04	UJ2	ELBOFLG	5516619	1.5	1.5	ELBOFLANGE 300# RF SCH 20/160 LTCS A350 LF2 CL.1	16	24	
2590	U04	UJ2	ELBOFLG	5516623	1.5	1.5	ELBOFLANGE 300# RF SCH XS/160 LTCS A350 LF2 CL.1	26	44	
2600	U04	UJ2	ELBOFLG	5516650	1.5	1.5	ELBOFLANGE 300# RF WT 14.0MM/160 LTCS A350 LF2 CL.1	46	48	

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - ALL SIZES. SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL:LTCS FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 232 °C COMMENT: REFER NOTE 800	12/16/2024	BY OZAC	U04	Page 4 of 7	150K21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
2610	U04	UJ2	ELBOFLG	5516682	1.5	1.5	ELBOFLANGE 300# RF WT 15.0MM/SCH 160 LTCS A350 LF2 CL.1	56	56	
2620										
2620	U04		ELBOW	5478701	0.75	1.5	ELL 90 DEG 3000# SW LTCS A350 LF2 CL.1	0	0	
2630	U04		ELBOW	5478702	0.75	1.5	ELL 45 DEG 3000# SW LTCS A350 LF2 CL.1	0	0	
2640										
2640	U04		TEE	5478703	0.75	1.5	TEE 3000# SW LTCS A350 LF2 CL.1	0	0	
2650										
2650	U04		CAP	5478704	0.75	1.5	CAP 3000# SW LTCS A350 LF2 CL.1	0	0	
2660	U04	UD6	CAP	5491312	0.75	0.75	CAP 3000# SCRD LTCS A350 LF2 CL.1	0	0	
2670										
2670			COUPLING	5478705	0.75	1.5	COUPLING 3000# SW LTCS A 350 LF2	0	0	
2680	U04		TEE RED	5478709	0.75	1	COUPLING RED 3000# SW LTCS A350 LF2 CL.1	1	1.5	
2690										
2690	U04		UNION	5478706	0.75	1.5	UNION 3000# SW LTCS A350 LF2 CL.1	0	0	
2700										
2700	U04		FITTING SCRD	5491303	0.5	0.75	PLUG ROUND HEAD SCRD LTCS A350 LF2 CL.1	0	0	
2710										
2710	U04		OLETS	5533072	0.75	1.5	SOCKOLET 3000# LTCS A350 LF2 CL.1	2	56	
2720	U04		OLETS	5533232	0.75	1.5	SOCKOLET FLAT BOTTOM 3000# LTCS A350 LF2 CL.1	0	0	
2730										
2730	U04	UD7	OLETS	5514772	0.75	0.75	THREDOLET 3000# LTCS A350 LF2 CL.1	2	56	
2740										
2740	U04	SC1	FLANGE	5599690	0.75	0.75	FLG SCRD (FNPT) 150# RF LTCS A350 LF2 CL.1	0	0	
2750	U04		FLANGE	5602303	0.75	1.5	FLG SW 150# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
2760	U04		FLANGE	5602356	0.75	1.5	FLG SW 300# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
2770	U04		FLANGE	5602484	0.75	0.75	FLG SW 600# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
2780										
2780		PJ9	CROSS	5416600	2	2	CROSS SCH XS LTCS A420 WPL6	0	0	
2790			CROSS	5416601	3	10	CROSS SCH STD LTCS ASTM A420 WPL6	0	0	
2800			CROSS	5416602	3	6	CROSS RED SCH STD LTCS ASTM A420 WPL6	4	8	
2810			CROSS	5416602	8	8	CROSS RED SCH STD LTCS ASTM A420 WPL6	10	10	
2820										
2820			NIPPLE	5541184	0.5	0.5	NIPPLE SCH XS/160 LTCS A333-6 POE/TOE 3" LG	0	0	
2830			NIPPLE	5541183	0.5	0.5	NIPPLE SCH XS/160 LTCS A333-6 POE/TOE 7" LG	0	0	
2840			NIPPLE	5541172	0.75	1.5	NIPPLE SCH 160 LTCS A333 GR.6 TBE 3" LONG	0	0	
2850			NIPPLE	5541173	0.75	1.5	NIPPLE SCH 160 LTCS A333 GR.6 TBE 7" LONG	0	0	
2860										
2860							***** SWAGE NIPPLES *****			
2870										
2870			RED CONC	5544132	0.75	1	SWAGE CONC SCH 160 LTCS A420 WPL6 TBE	1	1.5	
2880			RED CONC	5544094	0.75	1	SWAGE CONC SCH XS/160 LTCS A420 WPL6 PLE/TSE	1	1.5	
2890			RED CONC	5544135	0.75	1	SWAGE CONC SCH 160/XS LTCS A420 WPL6 TLE/PSE	1	1.5	
2900			RED CONC	5544084	0.75	1	SWAGE CONC SCH XS LTCS A420 WPL6 BLE/PSE	2	2	
2910			RED CONC	5544156	0.75	1.5	SWAGE CONC SCH STD/XS LTCS A420 WPL6 BLE/PSE	3	4	
2920			RED ECC	5549222	0.75	1	SWAGE ECC SCH 160 LTCS A420 WPL6 TBE	1	1.5	
2930			RED ECC	5549185	0.75	1	SWAGE ECC SCH XS/160 LTCS A420 WPL6 PLE/TSE	1	1.5	
2940			RED ECC	5549195	0.75	1	SWAGE ECC SCH 160/XS LTCS A420 WPL6 TLE/PSE	1	1.5	
2950			RED ECC	5549174	0.75	1.5	SWAGE ECC SCH XS LTCS A420 WPL6 BLE/PSE	2	2	
2960			RED ECC	5549156	0.75	1.5	SWAGE ECC SCH STD/XS LTCS A420 WPL6 BLE/PSE	3	4	
2970										
2970		FS1					***** FLANGES *****			
2980										
2980	U04	FA8	FLANGE	5599692	0.5	0.5	FLG SCRD (FNPT) REDUCING 150# RF LTCS A350 LF2 CL.1	0.75	1.5	
2990	U04	FA8	FLANGE	5599703	0.5	0.5	FLG SCRD (FNPT) REDUCING 300# RF LTCS A350 LF2 CL.1	0.75	0.75	
3000										
3000	U04		FLG ADAPTER	5851211	0.5	1	FLANGE ADAPTER 150# RF 316 SS X 6 MM TUBE	0	0	
3010		VC1	FLG ADAPTER	5851200	0.5	2	FLANGE ADAPTER 150# RF 316 SS X 12 MM TUBE	0	0	
3020	U04		FLG ADAPTER	5851203	0.75	1.5	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE	0	0	
3030		VC1	FLG ADAPTER	5851204	0.75	1.5	FLANGE ADAPTER 600# RF 316 SS X 12 MM TUBE	0	0	
3040										
3040	U04		FLANGE	5556264	0.5	24	FLG WN 150# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
3050	U04		FLANGE	5556261	2.5	14	FLG WN 150# RF LTCS A350 LF2 CL.1 SCH STD	0	0	
3060	U04		FLANGE	5556265	16	24	FLG WN 150# RF LTCS A350 LF2 CL.1 SCH 20	0	0	
3070	U04		FLANGE	5556278	16	24	FLG WN 150# RF LTCS A350 LF2 CL.1 WT 19.05MM	0	0	
3080	U04		FLANGE	5556266	26	44	FLG WN 150# RF LTCS A350 LF2 CL.1 SCH XS B16.47 SERIES A	0	0	
3090	U04	821	FLANGE	5556253	36	36	FLG WN 150# RF LTCS A350 LF2 CL.1 WT 22.2MM B16.47 SERIES A	0	0	
3100	U04	821	FLANGE	5556254	48	48	FLG WN 150# RF LTCS A350 LF2 CL.1 WT 25.4MM B16.47 SERIES A	0	0	
3110	U04		FLANGE	5556250	46	48	FLG WN 150# RF LTCS A350 LF2 CL.1 WT 14.0MM B16.47 SERIES A	0	0	
3120	U04		FLANGE	5556271	56	56	FLG WN 150# RF LTCS A350 LF2 CL.1 WT 15.0MM B16.47 SERIES A	0	0	
3130	U04		FLANGE	5556476	0.5	2	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
3140	U04		FLANGE	5556473	3	14	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH STD	0	0	
3150	U04		FLANGE	5556474	16	24	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH 20	0	0	
3160	U04		FLANGE	5556477	26	44	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH XS B16.47 SERIES A	0	0	

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - ALL SIZES. SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL:LTCS FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 232 °C COMMENT: REFER NOTE 800	12/16/2024	BY OZAC	U04	Page 5 of 7	150K21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
3170	U04	821	FLANGE	5556482	30	30	FLG WN 300# RF LTCS A350 LF2 CL.1 WT 19.05 MM B16.47 SERIES A	0	0	
3180	U04		FLANGE	5556450	46	48	FLG WN 300# RF LTCS A350 LF2 CL.1 WT 14.0MM B16.47 SERIES A	0	0	
3190	U04		FLANGE	5556478	56	56	FLG WN 300# RF LTCS A350 LF2 CL.1 WT 15.0MM B16.47 SERIES A	0	0	
3200	U04		FLANGE	5557036	0.5	2	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
3210	U04		FLANGE	5557033	3	12	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH STD	0	0	
3220	U04		FLANGE	5557285	16	24	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH 20	0	0	
3230	U04	817	FLANGE	5558036	1.5	1.5	FLG WN 600# RTJ LTCS A350 LF2 CL.1 SCH XS	0	0	
3240	U04	FS1	FLANGE	5556557	26	26	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH XS B16.47 SERIES A	0	0	
3250	U04	821	FLANGE	5557141	8	8	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
3260	U04	FS1	FLANGE	5555935	3	10	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH STD	0	0	
3270	U04	FS1	FLANGE	5557144	30	30	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH XS B16.47 SERIES A	0	0	
3280	U04	FS1	FLANGE	5557393	0.5	2	FLG WN 1500# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
3290	U04	821	FLANGE	5557397	2	2	FLG WN 1500# RF LTCS A350 LF2 CL.1 SCH 160 BORE	0	0	
3300	U04	FS1, 817	FLANGE	5557704	0.5	2	FLG WN 2500# RTJ LTCS A350 LF2 CL.1 SCH XS	0	0	
3310	U04		BLIND	5615566	0.5	24	FLG BLIND 150# RF LTCS A350 LF2 CL.1	0	0	
3320	U04		BLIND	5615570	26	56	FLG BLIND 150# RF LTCS A350 LF2 CL.1 B16.47 SERIES A	0	0	
3330	U04		BLIND	5615576	0.5	24	FLG BLIND 300# RF LTCS A350 LF2 CL.1	0	0	
3340	U04		BLIND	5615579	26	56	FLG BLIND 300# RF LTCS A350 LF2 CL.1 B16.47 SERIES A	0	0	
3350	U04	817		5615698	3	3	FLG BLIND 600# RTJ LTCS A350 LF2 CL.1	0	0	
3360	U04	817		5615614	2	2	FLG BLIND 1500# RTJ LTCS A350 LF2 CL.1	0	0	
3370										
3370							***** ORIFICE FLANGES *****			
3380										
3380	U04		ORIFICE FLG	5623947	2	2	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
3390	U04		ORIFICE FLG	5623944	3	14	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH STD	0	0	
3400	U04		ORIFICE FLG	5623941	16	24	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH 20	0	0	
3410	U04		ORIFICE FLG	5624800	26	44	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH XS ASME B16.47 SERIES A	0	0	
3420	U04		ORIFICE FLG	5624828	46	48	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 WT 14.0MM ASME B16.47 SERIES A	0	0	
3430	U04		ORIFICE FLG	5624804	56	56	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 15MM WT ASME B16.47 SERIES A	0	0	
3440										
3440							***** PIPE NIPPLES *****			
3450										
3450		810		0			NIPPLE PBE - CUT FROM PIPE			
3460										
3460		FS1					***** GASKETS *****			
3470										
3470			GASKET	5673043	0.5	24	GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
3480			GASKET	5674011	26	56	GASKET 150# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A	0	0	
3490			GASKET	5673033	0.5	24	GASKET 300# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
3500			GASKET	5674032	26	56	GASKET 300# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A	0	0	
3510	U04	GA6	GASKET	5673017	2	3	GASKET 300# 316SS KAMMPROFILE GRAPHITE ASME B16.5	0	0	
3520	U04		GASKET	5673103	0.5	24	GASKET 600# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
3530			GASKET	5674012	26	26	GASKET 600# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A	0	0	
3540			GASKET	5674012	30	36	GASKET 600# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A	0	0	
3550			GASKET	5673117	1.5	10	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
3560	U04		GASKET	5672251	0.5	2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
3570	U04	817		5674479	2	2	GASKET 1500# RTJ SS OCTAGONAL RING	0	0	
3580	U04	817	GASKET	5674480	0.5	2	GASKET 2500# RTJ SS OCTAGONAL RING	0	0	
3590	U04	817	GASKET	5673202	1.5	3	GASKET 600# RTJ SOFT IRON TEFLON COATED OCTAGONAL RING	0	0	
3600	U04		GASKET	5674033	30	30	GASKET 900# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A	0	0	
3610										
3610		B1					***** BOLTING *****			
3620										
3620	U04		BOLTING	5675770	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
3630										
3630		SB6	BOLTING	5675675	12	56	JACK SCREW HEX HEAD A320 GR L7M	0	0	
3640										
3640							***** BLINDS AND SPACERS *****			
3650										
3650			SPEC BLIND	5749580	0.75	14	SPECTACLE BLIND 150# RF LTCS A 516 GR.70	0	0	
3660			SPEC BLIND	5749581	1.5	10	SPECTACLE BLIND 300# RF LTCS A516 GR.70	0	0	
3670										
3670			PADDLE	5750023	16	24	PADDLE BLANK 150# RF LTCS A516 GR.70	0	0	
3680			PADDLE	5750026	26	56	PADDLE BLANK 150# RF LTCS A516 GR70 B16.47 SERIES A	0	0	
3690			PADDLE	5750100	12	24	PADDLE BLANK 300# RF LTCS A516 GR.70	0	0	
3700			PADDLE	5750028	26	56	PADDLE BLANK 300# RF LTCS A516 GR70 B16.47 SERIES A	0	0	
3710			PADDLE	5750240	16	24	PADDLE SPACER 150# RF LTCS A516 GR.70	0	0	
3720			PADDLE	5750196	26	56	PADDLE SPACER 150# RF LTCS A516 GR.70 B16.47 SERIES A	0	0	
3730			PADDLE	5750200	12	24	PADDLE SPACER 300# RF LTCS A516 GR.70	0	0	
3740			PADDLE	5750198	26	56	PADDLE SPACER 300# RF LTCS A516 GR70 B16.47 SERIES A	0	0	
3750										
3760										

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - ALL SIZES. SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL:LTCS FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 232 °C COMMENT: REFER NOTE 800	12/16/2024	BY OZAC	U04	Page 6 of 7	150K21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
3760							*****MISCELLANEOUS*****			
3770										
3770	U04		FLG GUARD	5558881	0.5	10	SAFETY SHIELD PTFE SUITABLE FOR 150# FLANGE	0	0	

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - ALL SIZES. SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL:LTCS FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 232 °C COMMENT: REFER NOTE 800	12/16/2024	BY OZAC	U04	Page 7 of 7	150K21